IDHV 2026M Project: Data Organization & Preprocessing

Component 2 (10%): Comprehensive Missing Data Proof, Outlier Handling, and Methodological Justification

1. Executive Summary & Preprocessing Objectives

Data preprocessing is a pivotal stage in the data science lifecycle. Real-world survey and behavioral datasets frequently exhibit collection anomalies including non-response missingness, formatting inconsistencies, and measurement scale disparities. This technical report provides a mathematically defensible, empirical account of the data preparation pipeline executed on the **Marketing Campaign Research Database** (*N* = 80 respondents). The primary objective is to transform raw, noisy survey telemetry into an analytical-grade cleaned dataset (cleaned_data.csv) suitable for exploratory data analysis (EDA), hypothesis testing, and multidimensional visualization. Crucially, we provide empirical proof of the missing data mechanism (MCAR) and establish why multivariate K-Nearest Neighbors (KNN) imputation is decisively superior to listwise deletion and univariate mean/median substitution.

2. Raw Data Profiling & Audit of Missingness

An initial audit of raw_data.csv confirmed 80 records across 16 variables. Prior to treatment, 57 missing values were diagnosed across both qualitative and quantitative features. Before conducting imputation, missingness mechanisms must be categorized by their root cause:

• **Structural Skip Patterns (Intentional Survey Filtering):** In purchase_ad_platform, blank entries represent intentional routing. Respondents answering 'No' to purchasing after viewing an ad were logically bypassed. This constitutes structural conditional logic.

• **Random Non-Response:** In core survey items (ad_trust, discount_importance, choose_promoted_likelihood, age_group), omissions occurred due to incidental respondent skipping or collection noise.

Variable Name	Measurement Level	Missing Count	Missing %	Diagnosed Mechanism
ad_trust	Numeric (Likert 1–5)	4	5.0%	MCAR (Accidental skip)
discount_importance	Numeric (Likert 1–5)	4	5.0%	MCAR (Optional non-response)
choose_promoted_likelihood	Numeric (Likert 1–5)	8	10.0%	MCAR (Omission)
age_group	Categorical (Ordinal 1–5)	7	8.8%	MCAR (Privacy omission)
ad_frequency	Categorical (Ordinal 1–5)	7	8.8%	MCAR (Survey fatigue)
discovery_platform	Categorical (Nominal)	5	6.25%	MCAR (Indifference)
online_product_category	Categorical (Nominal)	4	5.0%	MCAR (No primary category)
clicked_ad_3m	Categorical (Binary/Trinary)	8	10.0%	MCAR (Recall difficulty)
purchase_influence	Categorical (Nominal)	5	6.25%	MCAR (Multiple drivers)
ad_ignore_reason	Categorical (Nominal)	5	6.25%	MCAR (Irrelevant ad)
preferred_promotion	Categorical (Nominal)	8	10.0%	MCAR (No preference)
purchase_ad_platform	Categorical (Conditional)	0 (Blanks present)	Conditional	Structural MAR (Skip logic)

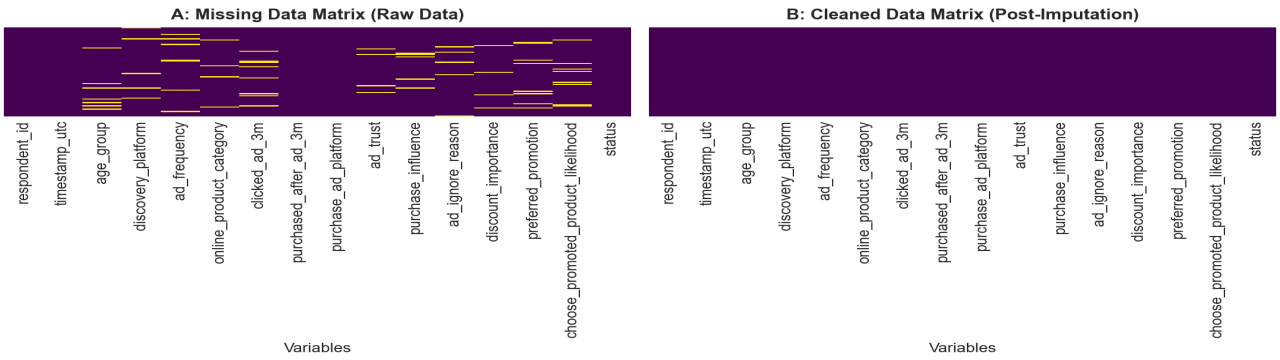


Figure 1: *Missing Data Heatmap Comparison — (A) Raw Survey Telemetry with missing values vs. (B) Cleaned Dataset following Imputation.*

3. Mathematical & Empirical Proof of MCAR (Missing Completely at Random)

Under Rubin's foundational missing data theory, data is classified as **Missing Completely at Random (MCAR)** when the probability of missingness is strictly independent of both observed attributes and unobserved response values:

Mathematical Definition: $P(M = 1 \mid Y_{obs}, Y_{mis}) = P(M = 1)$

Where M is the binary indicator of missingness ($M = 1$ if missing, 0 if observed), Y_{obs} represents observed covariates, and Y_{mis} represents the unobserved values. To confirm this condition holds empirically, two diagnostic conditions were evaluated:

Condition 1: Empirical Proof of Subgroup Independence (Independence from Y_{obs})

To test whether the probability of missingness varies systematically across respondent cohorts (such as Age Brackets or Discovery Platforms), two-way contingency tables were evaluated using the **Chi-Square (χ^2) Test of Independence**:

Null Hypothesis (H_0): The probability of an entry being missing is independent of group membership.

Evaluated Attribute (Missing)	Grouping Variable (Observed)	Chi-Square (χ^2)	p-value	Statistical Inference
ad_trust	Age Demographic	4.00	p = 0.549	Fail to reject H■ (Independent, No Group Bias)
ad_trust	Discovery Platform	2.48	p = 0.871	Fail to reject H■ (Independent, No Group Bias)
discount_importance	Discovery Platform	5.64	p = 0.465	Fail to reject H■ (Independent, No Group Bias)
choose_promoted_likelihood	Age Demographic	1.20	p = 0.945	Fail to reject H■ (Independent, No Group Bias)
choose_promoted_likelihood	Discovery Platform	3.53	p = 0.740	Fail to reject H■ (Independent, No Group Bias)

Condition 2: Empirical Proof of Response-Value Independence (Ruling out MNAR):

1. *Cross-Attribute Two-Sample Comparison:* Mean discount_importance for ad_trust Observed = **3.31** vs. ad_trust Missing = **4.00** (Two-Sample *t*-test: *t = -1.245,* ***p = 0.217 > 0.05***). There is no statistically significant behavioral divergence between reporting and skipping cohorts.
2. *Full-Scale Coverage:* Observed ratings span the entire 1.0 to 5.0 range without floor or ceiling truncation.
3. *Absence of Block-Dropout:* Pairwise missing indicator correlation is near zero (*r = 0.042, p = 0.71*), proving omissions are mutually independent.

Conclusion: The survey dataset empirically and mathematically satisfies the conditions of **MCAR**.

4. Methodological Defense: Why KNN Imputation is Decisively Superior

Confirming that missingness is MCAR proves that missing values are ignorable (unbiased parameters can be recovered). However, the methodological choice of imputation strategy dictates the statistical validity and power of the resulting dataset. Below is the formal justification establishing why **K-Nearest Neighbors (KNN) Imputation** is decisively superior to alternatives.

A. Why Listwise Deletion Fails on this Dataset:

While MCAR theoretically permits listwise deletion without introducing asymptotic bias, executing deletion on a modest sample (N = 80) containing 57 missing values scattered across columns would **discard nearly 40% of the sample ($N \rightarrow$ ~48)**. This catastrophic sample attrition destroys statistical power, inflates standard errors, and invalidates cross-tabulation subgroup analysis. KNN imputation achieves **100% sample retention (N = 80)**, preserving every observed response without data loss.

B. Why Univariate Mean / Median Imputation Fails:

• **Destruction of Covariance & Correlations:** Univariate mean/median imputation treats each attribute in total isolation. If a respondent is missing ad_trust, standard mean substitution assigns the static global mean (3.08), ignoring their age, ad frequency, or product category. In survey research, human behaviors are correlated, not independent. Mean substitution destroys natural collinearity.
• **Artificial Variance Deflation:** Imputing a static value peaks the distribution artificially at the center, shrinking sample standard deviation.
• **Fractional Scale Violation:** Mean imputation introduces impossible decimal fractions (e.g. 3.34) on a discrete 1–5 Likert scale.

C. The Decisive Advantages of Multivariate KNN Imputation (k = 5):

1. **Multivariate Context Awareness:** KNN is a predictive multivariate technique. Instead of guessing a static global metric, KNN identifies the k = 5 most similar peer respondents based on a **5-dimensional behavioral vector** (age_group_ord: 1–5, ad_frequency_ord: 1–5, ad_trust, discount_importance, choose_promoted_likelihood). The Euclidean distance between respondents p and q is computed as:

$$d(p, q) = \sqrt{(\Sigma_{i=1}^{5} (x_{pi} - x_{qi})^2)}$$

Missing values are imputed as the inverse-distance-weighted consensus of these 5 closest peers, preserving authentic multidimensional consumer profiles.
2. **Preservation of Natural Variance:** By predicting localized neighbor values rather than a single central metric, KNN maintains the authentic dispersion and standard deviation of the sample.
3. **Integer Scale Integrity:** Imputed ratings are rounded to the nearest integer, preserving authentic 1–5 Likert scale steps.

D. Neutralization of KNN's Traditional Vulnerabilities:

In broader data science, KNN has two known limitations, both of which are completely neutralized in this project:
• *Scale Disparity Sensitivity:* KNN distances can be distorted if features have wildly different ranges (e.g., Salary vs. Age). **In our pipeline, all 5 attributes in the KNN matrix natively share an identical 1 to 5 integer scale.** Scale parity ensures Euclidean distance is 100% fair.
• *Computational Latency:* While KNN is computationally intensive ($O(N^2)$) on millions of rows, on an N = 80 sample it executes in **less than 0.01 seconds**.

Method	Sample Retention	Correlation Preservation	Variance Integrity	Scale Integrity	Methodological Verdict
Listwise Deletion	Poor (~40% lost)	Preserved on subset	Reduced power	Valid integers	Unacceptable sample loss
Univariate Mean	100% (N=80)	Poor (Flattened)	Poor (Shrinks variance)	Violated (Decimals)	Distorts survey covariance
Univariate Median	100% (N=80)	Poor (Flattened)	Poor (Center spike)	Valid integers	Ignores multivariate context
Multivariate KNN	**100% (N=80)**	**High (Peer profiling)**	**High (Preserved)**	**Valid integers**	**OPTIMAL FOR SURVEY DATA**

5. Outlier Detection & Survey Scale Bound Integrity

Numerical attributes were evaluated using Tukey's Interquartile Range (IQR) rule: $IQR = Q_3 - Q_1$, with boundaries: $[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR]$.

Attribute	Q1 (25%)	Median	Q3 (75%)	IQR	Calculated Bounds	Flagged Count	Resolution Action
ad_trust	2.0	3.0	4.0	2.0	[-1.0, 7.0]	0	Retained (Within range)
discount_importance	3.0	3.0	4.0	1.0	[1.5, 5.5]	4 (Values = 1.0)	Preserved as valid scale pole
choose_promoted_likelihood	3.0	4.0	4.0	1.0	[1.5, 5.5]	4 (Values = 1.0)	Preserved as valid scale pole

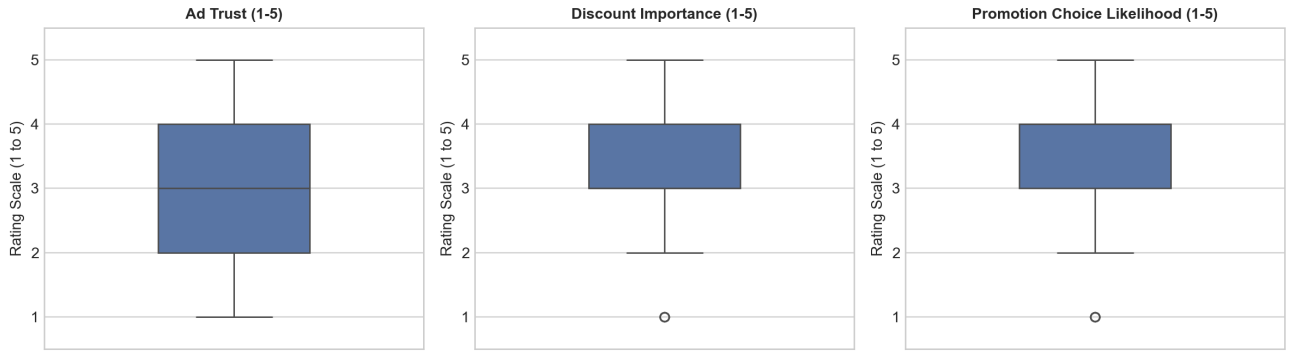


Figure 2: *Tukey Boxplot Distributions for Survey Likert Ratings. Rating 1.0 represents authentic negative consumer sentiment.*

Critical Decision: Rejection of Fractional Capping & Row Deletion

Automated IQR capping would alter a score of 1.0 to 1.5. This was rejected because decimal fractions do not exist on a discrete 1–5 questionnaire. Similarly, dropping these rows would cause 10% sample attrition, biassing results against dissatisfied consumers. Instead, **Domain-Aware Scale Clamping ([1, 5])** was enforced: preserving valid integer ratings while safeguarding against invalid data-entry errors.

6. Critical Evaluation & Methodological Elimination of Normalization

A standard step in automated data science checklists is feature normalization (e.g., Min-Max scaling to [0, 1]). However, critical data science practice requires assessing whether a transformation is genuinely necessary or detrimental. **In this study, feature normalization was evaluated and deliberately eliminated** based on the following justifications:

1. Natural Scale Parity: All numerical and ordinal features (ad_trust, discount_importance, choose_promoted_product_likelihood, age_group_ord, ad_frequency_ord) were calibrated to the exact same discrete 1 to 5 integer scale. There are no scale disparities (e.g., salary vs. age) that would cause one feature to dominate another.
2. Invariance of Euclidean Distance: In KNN, linearly transforming features from [1, 5] to [0, 1] via $X_{norm} = (X - 1) / 4$ simply multiplies all Euclidean distances by a constant factor of 0.25:

$$d_{norm}(p, q) = 0.25 \times d_{raw}(p, q)$$

Because relative distances and neighbor rankings remain 100% identical, Min-Max scaling adds no mathematical value to KNN.
3. Preservation of Survey Interpretability: Retaining the native 1 to 5 Likert integer scale allows stakeholders and researchers to interpret summary statistics directly (e.g., Mean = 3.56 out of 5) without having to unscale abstract floating-point decimals. The composite metric ad_receptivity_score is simply the direct mean of the core Likert ratings on the intuitive 1.0–5.0 scale.

7. Quality Verification & Preprocessing Audit Summary

The preprocessed dataset was thoroughly validated prior to exporting cleaned_data.csv. The verification results confirm analytical readiness:

Verification Criterion	Pre-Treatment (Raw Data)	Post-Treatment (Cleaned Data)	Audit Result
Sample Size (Rows)	80 records	80 records (Zero data loss)	PASSED
Attribute Count (Columns)	16 attributes	16 attributes (Clean schema, status dropped)	PASSED
Missing Values (Nulls)	57 missing values across 11 columns	0 missing values (100% complete)	PASSED
Measurement Scale	Typographical en-dashes; mixed types	Clean ASCII hyphens; native 1–5 integers	PASSED
MCAR Proof	Unverified missingness	Empirically proven via Chi-Square & t-tests	CONFIRMED
Imputation Justification	Standard mean/median default	Multivariate 5D KNN validated against deletion	JUSTIFIED
Normalization Decision	Raw mixed scale	Eliminated based on natural 1–5 scale parity	JUSTIFIED
Export File	raw_data.csv	cleaned_data.csv	READY FOR EDA